

Wentao Li

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Education

 **University of California, Berkeley** Berkeley, CA
VISITING STUDENT 2024.1 - 2024.5

- Course Paper: Optical Tests of Bell's Inequality (**full score**, given by Prof. Dan Stamper-Kurn of Berkeley Ultracold Atom group)
- Coursework: GPA **3.97/4**, Quantum Mechanics 2 (A+), Introduction to Quantum Computation (A), Nonlinear and Quantum Optics (A-)

 **Fudan University** Shanghai, China
BSc IN PHYSICS 2021.9 - 2026.6

- Coursework: GPA 3.44/4, Thermodynamics and Statistical Physics 1 (Honors, A), Thermodynamics and Statistical Physics 2 (A, top 5%), Computational Simulation of Physics (A-)

Research

 **Hong Kong University of Science and Technology, Guangzhou** Remote Collaboration
yao-rs OPEN SOURCE CONTRIBUTOR 2026.6 - now

- Re-write the Julia quantum computing software package yao.jl into the Rust package yao-rs to improve performance and reliability

 **University of California, Berkeley** Berkeley, CA
UNDERGRADUATE RESEARCHER, EHUD ALTMAN GROUP 2024.5 - now

- Obtained** the critical temperature and universality class of the decoherence-induced phase transition in the Kitaev honeycomb model via mean-field theory and frequency-shell renormalization group
- Mapped** the statistical entropy of a mixed state to the entanglement entropy of the Choi-Jamiolkowski (CJ) pure state exactly, thereby **revealed** the generic physical picture of entanglement formation between two CJ layers in decoherence-induced transitions
- Proposed** a fast tensor-network method for topological entanglement entropy based on the CJ isomorphism

 **Fudan University** Shanghai, China
COURSE PAPER, THERMODYNAMICS AND STATISTICAL PHYSICS 2 2024.2 - 2024.6

- Revealed** that dangerously irrelevant perturbations (DIP) are caused by the breakdown of the linear approximation for the renormalization group flow
- Heuristically justified** the superiority of the Monte Carlo Renormalization Group (MCRG) methods over existing analytic methods when applied to models with DIP's
- Interpreted** the results of MCRG numerics and **confirmed** that DIP's require a new set of critical exponents to describe

Professional Experiences

Xing-Ye Securities Shanghai, China
INTERN AT FIXED INCOME DEPARTMENT 2023.7 - 2023.8

- Analyzed monthly CPI commentary reports from various brokerage firms since August 2021; concluded that integrating the commodity basket with broader macroeconomic indicators and tailoring analytical frameworks are crucial for accurate qualitative CPI forecasting
- Streamlined quantitative CPI forecasting by optimizing Excel sheets to automate Wind database queries and real-time calculations; integrated automated error-handling to alert on missing or delayed National Bureau of Statistics (NBS) data

Projects and Societies

 **Swarma Club**
PARTNER CONTENT CREATOR

2026.3 – now

- Track the developments of interdisciplinary research of quantum technology and complex systems, and write blog posts or journal article explainers, such as *Q-day Crisis of Cryptocurrencies*, and *Quantum Reservoir Computation at the Edge of Chaos*

 **Quantum Computing Society of Berkeley**
ACTIVE MEMBER

Berkeley, CA

2024.2 - 2024.7

- Tracked the developments of quantum computing, and present in the journal club
- Gave a scientific talk on the potential applications of quantum computing to statistical physics

Skills and Interests

- Languages: English (IELTS: overall score 8.5), Chinese
- Programming: LaTeX, Typst, Python, Julia, Rust
- Interests: Strength training, self-hosting (<https://wentaoli.xyz>), cooking, coffee, photography (<https://unsplash.com/@vincenthetic>)